CP1.8: Proof by Mathematical Induction

1. The Structure of Induction

Mathematical induction proves a statement $P(n)$ is true for all integers $n \geq 1$ (or some starting value) through four steps:

1. **Base case:** Verify $P(1)$ is true directly.
2. **Inductive hypothesis:** Assume $P(k)$ is true for some arbitrary integer $k \geq 1$.
3. **Inductive step:** Using the assumption, prove $P(k+1)$ is also true.
4. **Conclusion:** State the result using the standard phrasing.

The conclusion MUST contain these elements (exact phrasing matters for marks):

"Since the statement is true for $n = 1$, and if it is true for $n = k$ then it is true for $n = k + 1$, by the principle of mathematical induction, [the statement] is true for all positive integers n."

Award-losing errors: Not including all three elements: (1) true for $n = 1$, (2) true for $k \Rightarrow$ true for $k + 1$, (3) "by mathematical induction" and the full conclusion statement.

2. Type 1: Summation Results

Proof: Prove by induction that $\displaystyle\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ for all positive integers n.

Base case ($n = 1$):

LHS: $\sum_{r=1}^{1} r = 1$

RHS: $\frac{1 \cdot 2}{2} = 1$

LHS = RHS, so the statement is true for $n = 1$. ✓

Inductive hypothesis:

Assume the statement is true for $n = k$, i.e., assume:

$$\sum_{r=1}^{k} r = \frac{k(k+1)}{2}$$

Inductive step: We need to show that $\sum_{r=1}^{k+1} r = \frac{(k+1)(k+2)}{2}$.

$$\sum_{r=1}^{k+1} r = \sum_{r=1}^{k} r + (k+1) = \frac{k(k+1)}{2} + (k+1) \quad \text{(using hypothesis)}$$

$$= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$

This is exactly the required result with $n = k + 1$. ✓

Conclusion:
The statement is true for $n = 1$. If it is true for $n = k$ then it is true for $n = k + 1$. By the principle of mathematical induction, $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ is true for all positive integers n.

Proof: Prove by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$.

Base case ($n = 1$): LHS $= 1$, RHS $= \frac{1 \cdot 2 \cdot 3}{6} = 1$. ✓

Hypothesis: Assume $\sum_{r=1}^{k} r^2 = \frac{k(k+1)(2k+1)}{6}$.

Step:

$$\sum_{r=1}^{k+1} r^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{k(k+1)(2k+1) + 6(k+1)^2}{6}$$

$$= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} = \frac{(k+1)(2k^2 + 7k + 6)}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$$

This equals $\frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$. ✓

Conclusion: By induction, true for all positive integers n.

3. Type 2: Divisibility

For divisibility proofs, assume the hypothesis and manipulate to show the expression for $k + 1$ is divisible by the required integer.

Proof: Prove by induction that $8^n - 1$ is divisible by 7 for all positive integers n.

Base case ($n = 1$): $8^1 - 1 = 7$, which is divisible by 7. ✓

Hypothesis: Assume $8^k - 1 = 7m$ for some integer m, i.e., $8^k = 7m + 1$.

Inductive step:
$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(7m + 1) - 1 = 56m + 8 - 1 = 56m + 7 = 7(8m + 1)$
Since m is an integer, $8m + 1$ is an integer. So $8^{k+1} - 1$ is divisible by 7. ✓

Conclusion: By mathematical induction, $8^n - 1$ is divisible by 7 for all positive integers n.

Proof: Prove by induction that $f(n) = 4^n + 6n - 1$ is divisible by 9 for all positive integers n.

Base case ($n = 1$): $f(1) = 4 + 6 - 1 = 9$. Divisible by 9. ✓

Hypothesis: Assume $4^k + 6k - 1 = 9m$ for some integer m, so $4^k = 9m - 6k + 1$.

Inductive step: $f(k + 1) = 4^{k+1} + 6(k + 1) - 1 = 4 \cdot 4^k + 6k + 5$
$= 4(9m - 6k + 1) + 6k + 5 = 36m - 24k + 4 + 6k + 5 = 36m - 18k + 9$
$= 9(4m - 2k + 1)$
Since $4m - 2k + 1$ is an integer, $f(k + 1)$ is divisible by 9. ✓

Conclusion: By induction, $f(n)$ is divisible by 9 for all positive integers n.

Exam tip: For divisibility proofs, always introduce a variable (usually m) for the integer. Write $4^k = 9m - 6k + 1$ (or similar) and then substitute. Do NOT just write "therefore divisible" without showing all the algebra.

4. Type 3: Matrix Powers

Proof: Prove by induction that $M^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ where $M = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

Base case ($n = 1$): $M^1 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2(1) \\ 0 & 1 \end{pmatrix}$ ✓

Hypothesis: Assume $M^k = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$.

Inductive step:

$$M^{k+1} = M^k \cdot M = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \cdot 1 + 2k \cdot 0 & 1 \cdot 2 + 2k \cdot 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 + 2k \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2(k+1) \\ 0 & 1 \end{pmatrix}$$

This matches the formula with $n = k + 1$. ✓

Conclusion: By mathematical induction, $M^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ for all positive integers n.

Example 2: Prove that $A^n = \begin{pmatrix} 3^n & 0 \\ 3^n - 1 & 1 \end{pmatrix}$ where $A = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$.

Base case ($n = 1$): $A^1 = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$. RHS: $\begin{pmatrix} 3 & 0 \\ 3 - 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$. ✓

Hypothesis: Assume $A^k = \begin{pmatrix} 3^k & 0 \\ 3^k - 1 & 1 \end{pmatrix}$.

Step:

$$A^{k+1} = A^k \cdot A = \begin{pmatrix} 3^k & 0 \\ 3^k - 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \cdot 3^k & 0 \\ 3(3^k - 1) + 2 & 1 \end{pmatrix} = \begin{pmatrix} 3^{k+1} & 0 \\ 3^{k+1} - 1 & 1 \end{pmatrix}$$

This is the formula with $n = k + 1$. ✓

Exam tip: For matrix induction, use $M^{k+1} = M^k \cdot M$ (multiply the hypothesis by M on the right). Explicitly state you are using the inductive hypothesis when substituting the assumed form.

5. Type 4: Recurrence Relations

Proof: A sequence is defined by $u_1 = 3$, $u_{n+1} = 3u_n - 4$. Prove by induction that $u_n = 2 \cdot 3^{n-1} + 2$.

Base case ($n = 1$): $u_1 = 2 \cdot 3^0 + 2 = 2 + 2 = 4$... wait, we're given $u_1 = 3$.
Check: $2 \cdot 3^{1-1} + 2 = 2(1) + 2 = 4 \neq 3$.
This formula would be wrong in a real question — the example below uses a correct formula.

Correct example: $u_1 = 5$, $u_{n+1} = 2u_n - 1$. Prove $u_n = 3 \cdot 2^{n-1} + 1$.

Base case ($n = 1$): $u_1 = 3 \cdot 2^0 + 1 = 3 + 1 = 4$.
But $u_1 = 5$. So check formula: $3 \cdot 1 + 1 = 4 \neq 5$. Let's use: $u_n = 4 \cdot 2^{n-1} + 1$.
$n = 1$: $4 + 1 = 5$ ✓

Hypothesis: Assume $u_k = 4 \cdot 2^{k-1} + 1$.

Step: $u_{k+1} = 2u_k - 1 = 2(4 \cdot 2^{k-1} + 1) - 1 = 4 \cdot 2^k + 2 - 1 = 4 \cdot 2^k + 1$
This equals $4 \cdot 2^{(k+1)-1} + 1$. ✓

Conclusion: By induction, $u_n = 4 \cdot 2^{n-1} + 1$ for all positive integers n.

6. Common Mistakes and Mark-Scheme Notes

Mistake	Correction
Not clearly stating the inductive hypothesis as an assumption	Write "Assume that [formula] for some positive integer k"

Not showing $n = 1$ satisfies both sides independently	Compute LHS and RHS separately, then confirm they're equal
Circular reasoning: using the $k + 1$ result before proving it	Start from the k result, then derive the $k + 1$ result
Incomplete conclusion	Must state: "true for $n = 1$", "true for k implies true for $k + 1$", "by mathematical induction", full conclusion
Not factoring out correctly in divisibility proofs	Show the expression equals the divisor times an integer explicitly
Matrix: multiplying in wrong order	$M^{k+1} = M^k \cdot M$ (multiply on the right)

Full conclusion template (memorise this):
"Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$ for all positive integers k, by the principle of mathematical induction, [state the result, e.g., $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$] is true for all positive integers n."

The Edexcel mark scheme typically awards 1 mark for the correct base case, 1 mark for correctly assuming the hypothesis, 2–3 marks for the inductive step, and 1 mark for a complete conclusion. Skimping on the conclusion wastes a mark that is very easy to obtain.